Problem 16

Evaluate the integral:

$$\int (2x - 3\sin x + \cos x)\, dx.$$

Solution

We are tasked with solving the integral:

$$\int (2x - 3\sin x + \cos x)\, dx.$$

The integral can be separated into three simpler integrals:

$$\int (2x - 3\sin x + \cos x)\, dx = \int 2x\, dx - \int 3\sin x\, dx + \int \cos x\, dx.$$

Step 1: Evaluate each term

1. **First term: $\int 2x\, dx$:** Using the power rule for integration:

$$\int x^n\, dx = \frac{x^{n+1}}{n+1} \quad (\text{for } n \neq -1),$$

we have:

$$\int 2x\, dx = 2 \cdot \frac{x^2}{2} = x^2.$$

2. **Second term: $\int -3\sin x\, dx$:** Using the fact that the integral of $\sin x$ is $-\cos x$:

$$\int -3\sin x\, dx = -3 \int \sin x\, dx = -3(-\cos x) = 3\cos x.$$

3. **Third term: $\int \cos x\, dx$:** Using the fact that the integral of $\cos x$ is $\sin x$:

$$\int \cos x\, dx = \sin x.$$

Step 2: Combine the results

Combine all three terms:

$$\int (2x - 3\sin x + \cos x)\, dx = x^2 + 3\cos x + \sin x + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$\int (2x - 3\sin x + \cos x)\, dx = x^2 + 3\cos x + \sin x + C.$$